1. Executive Summary

Northwind Analytics deployed a retrieval-augmented question answering system across its internal documentation in the third quarter. The programme covered 14,200 documents drawn from engineering runbooks, customer support transcripts and the compliance handbook.

Average time to locate an authoritative answer fell from 8.4 minutes to 41 seconds. Support ticket deflection improved by 23 percent. The total infrastructure cost was 1,180 US dollars per month, of which 62 percent was inference and 38 percent was vector storage.

The board approved a phase two rollout to the field sales organisation, scheduled to begin in January.

2. System Architecture

Documents enter the pipeline through a watched object store bucket. Each file is parsed, normalised and split into overlapping windows of roughly 1,000 characters with 150 characters of overlap. Overlap matters because a sentence split across a boundary otherwise becomes unretrievable.

Chunks are embedded with the all-MiniLM-L6-v2 sentence transformer, which produces 384-dimensional vectors. Vectors are written to a ChromaDB collection configured for cosine distance.

At query time the user question is embedded with the same model. The five nearest chunks are retrieved, filtered by a distance threshold, and assembled into a numbered context block that is passed to the language model together with a grounding instruction.

3. Evaluation Results

The team evaluated the system against a held-out set of 480 questions written by domain experts. Answer accuracy, judged by human raters, reached 91.2 percent. Retrieval recall at five was 88.7 percent.

Hallucination rate, defined as any factual claim not supported by the retrieved context, was measured at 2.1 percent, down from 19.6 percent for the same model prompted without retrieval. This single figure was the strongest argument for the programme's continued funding.

Median end-to-end latency was 1.9 seconds at the ninety-fifth percentile, well inside the three-second target agreed with the support organisation.

4. Risks and Limitations

Scanned documents without a text layer are not indexed. An optical character recognition stage is planned but not yet funded.

The system does not currently reconcile contradictory sources. When two documents disagree, the answer reflects whichever chunk ranked higher, and the contradiction is not surfaced to the reader.

Access control is enforced at the collection level rather than the document level. Until per-document filtering ships, the compliance handbook is indexed in a separate collection restricted to the legal team.

5. Recommendations

First, fund the optical character recognition stage. Roughly 6 percent of the corpus is currently invisible to search.

Second, add a contradiction detector that flags when retrieved chunks disagree, rather than silently preferring the top-ranked one.

Third, migrate access control to per-document metadata filtering so that a single collection can serve every team.

Fourth, re-evaluate the embedding model each quarter. The field moves quickly and a better model is the cheapest available accuracy gain.